Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 29)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 The Rio Bravo Conservation and Management Area in Belize is a largely undamaged rainforest with over 200 species of tree, 400 species of bird and 70 species of mammal. It is managed by a Belizean NGO to conserve wildlife, produce resources, sequester carbon and benefit the local economy.

0 1 . 1 Explain why creating the reserve could have threatened local communities, and describe how the project avoids this.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 1 . 2 Explain **four** of the rules followed when trees are selected and felled that keep the exploitation sustainable.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 1 . 3 Each felled tree is given a unique identification tag. Explain how this supports Forest Stewardship Council accreditation.

[2 marks]

.....

.....

.....

Do not write outside the box

Do not write outside the box

0 1 . 4 Describe **three** ways the reserve raises money without damaging the forest.

[3 marks]

0 1 . 5 Explain why selective logging of this kind causes much less ecological damage than clear felling.

[3 marks]

15

0 2 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

0 2 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

5

0 3 . 1 Which of the following habitats has low biodiversity but a high proportion of endemic species? Tick one box.

[1 mark]

A Tropical rainforest ☐

B Oceanic islands ☐

C Temperate broadleaf woodland ☐

D Mangroves ☐

0 3 . 2 Compare the ecological features of tropical rainforest and Antarctica that explain their very different biodiversity.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

5

Do not write outside the box

0 4 The Isle of Eigg (area 31 km^2, population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

Table 1

Source	Capacity / kW
Hydroelectric (3 generators)	110
Solar PV	50
Wind (4 turbines)	24
Diesel generators (back-up)	160

0 4 . 1 Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 4 . 2 Describe **three** other ways in which the Eigg community has increased its sustainability.

[3 marks]

1 .....

.....

2 .....

.....

3 .....

.....

0 4 . 3 Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

[2 marks]

.....

.....

.....

Do not write outside the box

Do not write outside the box

.....

0 4 . 4 Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

.....

.....

.....

.....

10

0 5 Table 1 lists some feedback mechanisms that affect the Earth's climate.

0 5 . 1 Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	**Type of feedback**	
Higher temperatures increase cloud cover, raising albedo		
Melting permafrost releases methane		
Higher CO_2 levels increase photosynthesis and carbon sequestration		
Melting ice reduces albedo		

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 5 . 2 Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

Do not write outside the box

0 5 . 3 Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

0 5 . 4 Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

10

0 6 . 1 Explain **four** properties of water that make it essential for life.

[4 marks]

0 6 . 2 Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

0 6 . 3 Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

.....

.....

.....

.....

10

0 7 Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 7 . 1 Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

.....

Do not write outside the box

Do not write outside the box

0 7 . 2 Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

0 7 . 3 Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

0 7 . 4 Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

10

0 8 . 1 Which of the following best describes a **representative** sample? Tick one box.

[1 mark]

A A sample taken from the most convenient location ☐

B A sub-sample that accurately reflects the complete data set ☐

C The largest sample that can be collected ☐

D A sample that supports the hypothesis ☐

0 8 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow.

[2 marks]

.....

.....

.....

.....

0 8 . 3 Explain why deliberately choosing sampling sites introduces bias.

[2 marks]

.....

.....

.....

.....

5

0 9 Table 1 shows food conversion ratios (FCR) for livestock: the mass of feed needed to produce 1 kg of live mass gain.

Table 1

Livestock	FCR / kg feed per kg gain
Farmed salmon	1.2
Chicken	1.8
Pork	3.0
Beef (feedlot)	7.0
Sheep	8.0

0 9 . 1 Use Table 1 to explain why intensive poultry and fish production convert feed to food more efficiently than beef.

[3 marks]

.....

Do not write outside the box

Do not write outside the box

0 9 . 2 Explain **three** ways farmers reduce food chain energy losses in livestock production.

[3 marks]

0 9 . 3 Explain why growing crops for direct human consumption usually produces more food per hectare than rearing livestock, and describe **one** situation where livestock is the better option.

[4 marks]

10

1 0 . 1 Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

.....

.....

.....

.....

1 0 . 2 Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

.....

.....

.....

.....

1 0 . 3 Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

.....

.....

.....

.....

1 0 . 4 Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

Do not write outside the box

Extra space

.....

.....

.....

.....

.....

.....

15

Do not write outside the box

Write an essay on one of the following topics.

11 . 1 Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

OR

11 . 2 Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

Do not write outside the box

END OF QUESTIONS